

Privacy-Preserving EEG Classification with Fuzzy Unlearning

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A symbolic illustration of EEG brain activity with digital waveforms. Image source: [12].

Abstract

Machine learning models developed on brain-computer interface data pose significant privacy concerns because electroencephalographic signals contain subject-specific patterns that support biometric identification. Under the General Data Protection Regulation, individuals have the right to request the deletion of their personal data from systems in use. This necessitates subject-level unlearning methods for EEG-based classification. This paper presents an end-to-end pipeline for privacy-preserving EEG classification that employs fuzzy unlearning and studies the trade-off between retained utility and the reduced identifiability of an erased subject. The study uses a small multi-subject EEG dataset containing short recordings from 19 conventional channels collected during a cognitive task. The signals are segmented into sliding windows and transformed into tabular feature vectors using statistical descriptors and spectral-domain band-power features from the delta, theta, alpha, beta, and gamma bands. A Random Forest classifier is evaluated in three settings: a baseline model trained on all subjects, exact

unlearning through retraining after removing the target subject, and fuzzy unlearning through similarity-based sample reweighting. The results show clear differences between the three settings. The baseline model sustains strong subject-specific information and achieves a retained subject accuracy of 0.8971, while also showing near-perfect training performance, suggesting overfitting. Exact retraining provides the strongest balance between privacy-oriented forgetting and retained utility, achieving 0.9029 accuracy and 0.8998 macro F1 on the remaining subjects while producing substantially more uncertain predictions on samples from the forgotten subject. Fuzzy unlearning also reduces the learnability of the forgotten subject and moves model behavior toward exact retraining, but with a modest utility loss, reaching 0.8800 retained accuracy and 0.8719 macro F1. The study also exposes important pragmatic limitations. Suitable EEG datasets for subject-level unlearning remain scarce; sliding-window segmentation yields highly correlated samples, increasing the risk of overfitting; and forgetting after subject removal cannot be evaluated solely by accuracy. Instead, it requires confidence and uncertainty-based analysis. Overall, the findings show

that EEG unlearning is feasible in a controlled feature-based setting and that fuzzy unlearning is a likely approximate strategy, although its effectiveness still depends strongly on the dataset scale, sample dependence, and model design.

CCS Concepts

• **Security and privacy** → **Privacy protections; Security and privacy in machine learning; • Computing methodologies** → *Supervised learning; Ensemble methods*; • **Applied computing** → Health care information systems; Life and medical sciences.

Keywords

machine unlearning, electroencephalography, EEG biometrics, fuzzy unlearning, privacy-preserving machine learning, subject-level unlearning, biometric privacy

1 Introduction

The worldwide adoption of digital services and data-based technologies has intensified concerns about the protection of personal information. The General Data Protection Regulation (GDPR) establishes strict safeguards for individuals whose data are collected, stored, and processed online [1]. One of the main protections regarding this regulation is the right to erasure, often referred to as the right to be forgotten, which permits individuals to request removal of their personal data when the first purpose no longer applies, when consent has been withdrawn, or when processing was unlawful. In machine learning systems, satisfying this requirement is not straightforward because personal data may remain embedded in trained models even after the underlying records have been removed from storage [5]. This challenge is especially important for electroencephalography (EEG) data. EEG signals are not only physiological measurements, but also highly sensitive biometric and cognitive data. Prior work has shown that brain signals can reveal rich information about a person’s intentions, mental states, affective reactions, memory-related processes, and neurological conditions [3]. Since EEG directly captures electrical activity from the brain, it can also reveal subject-specific patterns that individuals cannot easily control or conceal. From a security and privacy perspective, this creates clear risks of profiling, health-related inference, and persistent identification using neural signatures. As a result, EEG-based machine learning systems should be evaluated not only for forecasting performance, but also for how much sensitive information they retain about the individuals whose data were used for training [3]. This privacy problem becomes more difficult under the GDPR when EEG data are used in machine learning. Personal data may remain embedded in a trained model even after the original records have been deleted, and the model may still be influenced by them in later predictions [5, 8]. This is especially relevant for EEG classification, where subject-specific brain patterns may continue to affect the model even after the corresponding recordings are removed [11]. Machine unlearning is a promising approach to addressing this problem. Its goal is to remove the influence of a single person from a trained model whilst preserving the model’s usefulness for the remaining data [8]. Exact retraining does this by retraining the model without the target subject, but it can be expensive. Therefore, approximate methods are also important. In

this work, fuzzy unlearning is used as a practical approximation by reducing the influence of samples most similar to the forgotten subject. This paper studies privacy-preserving EEG classification using a subject-level unlearning pipeline. The aim is not only to measure classification reliability, but also to examine whether a trained EEG model still retains reliable subject-specific information after a deletion request. The pipeline uses sliding window EEG segmentation, statistical and frequency features, and a Random Forest classifier. Three settings are evaluated: a baseline model trained on all subjects, exact unlearning through retraining after removing the target subject, and fuzzy unlearning through reduced weighting of samples most similar to the forgotten subject. This design enables examination of the privacy-utility trade-off under realistic EEG data parameters, where limited dataset size and high correlations among windowed samples make evaluation more challenging. To guide the study, the following research questions are addressed:

- **RQ1:** How strongly does the baseline EEG model retain subject-specific information?
- **RQ2:** How effectively does exact retraining remove the forgotten subject while protecting utility?
- **RQ3:** Can fuzzy unlearning approximate exact retraining in balancing privacy and utility?
- **RQ4:** How do limited dataset size and sample correlation affect the interpretation and reliability of the observed results?

The main contribution of this work is an end-to-end evaluation of EEG subject unlearning from a privacy and security perspective. The study shows how strongly subject-specific information can persist in an EEG classifier, how exact retraining serves as an accepted benchmark for removal, and how fuzzy unlearning delivers a practical approximation under realistic data boundaries.

2 Related Work

Research on EEG has consistently shown that brain signals contain rich information about both transient mental states and stable personal characteristics. One important line of research has focused on EEG-based recognition of affective and cognitive conditions. Wang et al. demonstrated that EEG recordings can be transformed into features that reflect emotional and mental states under an affective movie induction protocol [13]. In their study, multiple feature types, including power spectral, wavelet, and nonlinear dynamical features, were extracted from EEG signals and used to train machine learning models for classifying emotional states [13]. This work is relevant because it demonstrates that EEG signals not only capture task-related neural activity but also encode detailed internal information about an individual. From a privacy perspective, this is significant, as signals that reveal emotional or mental states are considered highly sensitive under data protection frameworks such as the GDPR [1]. Beyond mental-state analysis, EEG has also been investigated as a biometric modality for user identification and authentication. In contrast to traditional biometrics, including fingerprints or face images, EEG-based biometrics rely on neural activity patterns that are difficult to imitate and can reflect subject-specific characteristics at a deeper physiological level. Mao et al. proposed a deep learning method for EEG-based biometric

identification. They showed that EEG signals contain stable, subject-specific information that supports reliable identification across a relatively large population of subjects [7]. Their results are important for the present study because they strengthen the claim that trained EEG classifiers can retain subject-identifiable information, even when the system’s original goal is not explicitly biometric authentication. In other words, once a machine learning model is trained on EEG data, it may implicitly memorize subject-specific neural signatures that remain useful for identification. Another privacy concern is that EEG biometrics may reveal more than identity alone. Uhl and Holler showed that EEG-based representations can leak sensitive attributes such as sex, age, neurological condition, and medication status [4]. This finding is highly relevant for privacy-protecting EEG classification, since it suggests that EEG models can leak information beyond user recognition. A deployed EEG model may expose health or demographic information that the subject did not intend to share. As a result, the security perspective broadens: the goal becomes not only preventing reidentification but also reducing residual neural data that carries broader sensitive information. This research collectively provides the foundation for this paper. First, EEG signals encode affective and cognitive state information [13]. Second, they support subject-level identification via machine learning [7]. Third, EEG-based representations can leak private attributes beyond identity [4]. This literature shows that EEG data are highly informative and sensitive, making them important for studying privacy-preserving machine learning and subject-level data removal. Alongside EEG research, the machine learning field has explored machine unlearning, which removes selected data influences from trained models without full retraining. This effort connects to the right to erasure under the GDPR, as deleting raw records alone is insufficient if their effects remain embedded [8]. Shaik et al. surveyed machine unlearning [10], categorizing the literature and highlighting challenges such as efficiency, evaluation, standardization, interpretability, and the balance between privacy and utility [10]. Their survey situates this study within the broader landscape of unlearning, given that exact retraining is the gold standard, yet approximate methods are appealing when retraining is impractical. Among approximate methods and flexible strategies have become increasingly relevant in settings where privacy requirements must be balanced against efficiency constraints. Qu and Kong proposed a fuzzified federated multi-task unlearning framework for swarm consumer electronics systems [9]. Their method uses fuzzy logic to model data sensitivity and the strength of an unlearning request, facilitating selective removal of data influence without full retraining of the global system [9]. Although their application domain differs from EEG classification, the conceptual contribution is highly relevant to the present work. It shows that fuzzy reasoning can be used to control the degree of forgetting in a flexible, computationally optimized manner. This idea motivates the use of fuzzy unlearning in the current EEG pipeline, intending to reduce the influence of retained samples most similar to the forgotten subject. The relation between exact retraining and approximate unlearning is particularly important for privacy-protecting EEG systems. Exact retraining provides the clearest practical reference, as it rebuilds the model as if the forgotten subject had never contributed to the training data. However, exact retraining can be computationally costly and may be difficult

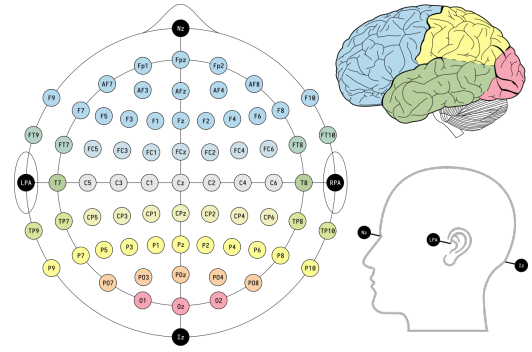


Figure 1: EEG electrode placement according to the 10–10 system with additional anatomical information. The 19 channels used in this study, Fp1, Fp2, F3, F4, F7, F8, T3, T4, C3, C4, T5, T6, P3, P4, O1, O2, Fz, Cz, and Pz, correspond to a subset of this standard layout. Figure from Wikimedia Commons by Laurens R. Krol, CC0 [6].

to apply repeatedly in actual systems. Approximate unlearning methods aim to mimic the behavior of exact retraining while reducing computational overhead. From an EEG-based classification perspective, this is especially valuable because EEG data sets are often limited in size, preprocessing is nontrivial, and the resulting feature vectors can still preserve subject-specific signatures. This makes EEG a challenging but meaningful domain in which to study whether fuzzy unlearning can provide a realistic privacy-preserving alternative. Despite the relevance of these individual research areas, a clear gap remains at their intersection. Existing EEG studies have established the sensitivity of neural signals and their value for subject identification [4, 7, 13]. Existing unlearning research has developed methods to remove data influence from trained models and has stressed the growing need for privacy-preserving deletion mechanisms [8–10]. However, relatively little work has directly connected subject-level EEG classification with machine unlearning in a security and privacy context. In particular, there is limited work that compares exact retraining and fuzzy unlearning with respect to the utility and preservation and reduction of residual subject identifiability in EEG-based systems. This gap motivates the present study.

3 Methodology

3.1 Threat Model and Problem Description

This paper studies a subject-level unlearning setting for EEG classification. One subject is selected as the target to forget, while the remaining subjects form the retain set. Three model settings are compared: a baseline model trained on all subjects, exact unlearning through retraining without the target subject, and fuzzy unlearning through similarity-based sample reweighting. The goal is to reduce subject-specific information related to the forgotten subject while preserving classification performance on the remaining subjects. This setup is consistent with the general machine unlearning objective of impairing performance on the forgotten data while maintaining satisfactory utility on the remaining data [11]. The broader privacy, utility, and efficiency motivations for approximate

unlearning are similarly aligned with recent fuzzy unlearning work [9].

3.2 Dataset Description

The experiments use an EEG dataset recorded before and during a mental arithmetic task. The recordings were acquired monopolarly with the Neurocom EEG 23-channel system (XAI-MEDICA, Ukraine). Silver/silver chloride electrodes were placed according to the international 10/20 system and referenced to interconnected ear electrodes. During acquisition, a high-pass filter with a 30 Hz cut-off frequency and a 50 Hz power line notch filter were applied. The original set of recordings was preprocessed using Independent Component Analysis to remove artifacts from eye movements, muscle activity, and cardiac interference. Each recording is an artifact-free EEG segment of 60 seconds. The task consisted of serial subtraction of a two-digit number from a four-digit number after oral presentation of the numbers. For the present study, the dataset contains 36 subject files in CSV format, converted from the original data EDF recordings. Each file contains 19 EEG channels: *Fp1*, *Fp2*, *F3*, *F4*, *F7*, *F8*, *T3*, *T4*, *C3*, *C4*, *T5*, *T6*, *P3*, *P4*, *O1*, *O2*, *Fz*, *Cz*, *Pz*. These channels serve as input variables for the classification pipeline. The dataset is sourced from the publicly available mental arithmetic EEG dataset described in [2, 14].

3.3 Data cleaning and Data Preparation

Each CSV file corresponds to one subject and is loaded as a multi-channel EEG recording. Since the CSV files do not contain a header row, all files are read with explicit header-free loading and then assigned the 19 standard channel names listed above. Only the 19 selected EEG channels are retained for further processing. To estimate the sampling frequency used in the analysis, divide the number of recorded samples in a 60-second segment by the recording duration. This estimated sampling rate is then used in the spectral feature extraction stage. For each subject, the raw EEG matrix is split chronologically into training and test portions using an 80:20 ratio. This subject-wise split ensures that the test data comprises later-unseen segments of the same recording and avoids mixing train and test windows within the same time range.

3.4 Sliding Window Segmentation

To transform the continuous EEG recordings into samples suitable for machine learning, each subject recording is segmented into fixed-length sliding windows. Let $x \in \mathbb{R}^{T \times C}$ denote the multichannel EEG recording of one subject, where T is the number of time samples and $C = 19$ is the number of channels. A window of length L samples is moved across the signal with a predefined step size S , producing a sequence of overlapping windows:

$$x^{(1)}, x^{(2)}, \dots, x^{(N)}, \quad x^{(i)} \in \mathbb{R}^{L \times C}.$$

In the implementation, the same window length is used for all subjects. Overlapping windows are used for the training set to increase the number of training samples, while a non-overlapping configuration is used for the test data to reduce redundancy during evaluation. Similar overlap-based segmentation strategies are commonly used in EEG learning pipelines to increase the number of training samples from limited recordings [11]. This design provides

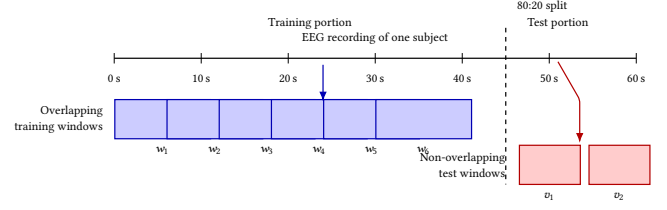


Figure 2: Illustration of the sliding window segmentation strategy. Each subject recording is first split chronologically into training and test portions. Overlapping windows are generated from the training portion to increase the number of samples, while non-overlapping windows are used on the test portion to reduce redundancy during evaluation.

a practical balance between sample generation and fairer testing. At the same time, it introduces correlation between nearby windows, which is an important limitation discussed later in the paper.

3.5 Feature Extraction

For each EEG window, a feature vector is extracted by combining time-domain and frequency-domain descriptors for every channel. The feature design uses a handcrafted tabular representation rather than direct end-to-end deep learning, because the goal is to study unlearning behavior in a controlled and interpretable setting.

For each channel within a window, five statistical features are computed:

$$\mu, \sigma, \text{var}, \text{min}, \text{max}$$

where μ is the mean, σ is the standard deviation, and var is the variance of the signal amplitude inside the window.

In addition, spectral band power features are computed using Welch's method. For each channel, the power spectral density is integrated over the standard EEG bands:

$$\begin{aligned} \delta &: 0.5\text{--}4 \text{ Hz}, \\ \theta &: 4\text{--}8 \text{ Hz}, \\ \alpha &: 8\text{--}13 \text{ Hz}, \\ \beta &: 13\text{--}30 \text{ Hz}, \\ \gamma &: 30\text{--}45 \text{ Hz}. \end{aligned}$$

The final feature vector for each window is formed by concatenating all statistical and band-power features across the 19 channels. These feature vectors form the tabular input to the classification and unlearning pipeline.

3.6 Baseline Classification Model

The baseline model serves as the reference system before any unlearning is applied. Its purpose is to measure how strongly the classifier retains subject-specific information when trained on the full dataset, including both the retained subjects and the target subject to forget. This provides the comparison point for evaluating the effect of exact and fuzzy unlearning. A Random Forest classifier is used as the baseline model. The model is trained on the complete baseline training set after feature extraction and standardization. In this setting, the classifier learns to predict all subject classes, including the forgotten subject. As a result, strong performance on

forgotten-subject test samples indicates that subject-specific information is still present in the trained model. The baseline model is evaluated on two subsets. First, it is tested on the retained-subject test set to measure standard classification utility. Second, it is tested on the forgotten-subject test set to examine whether the model continues to recognize the subject that should later be removed. Since the baseline classifier still includes the forgotten subject as a valid class, its predictions on forgotten-subject samples provide a direct indication of how much subject-specific information is encoded before unlearning. For this baseline setting, a StandardScaler is fitted on the full baseline training set, which includes both the retained subjects and the target subject to forget. The same transformation is then applied to the corresponding retained-subject and forgotten-subject test sets. This ensures consistent preprocessing for the reference model before comparison with the unlearning variants.

3.7 Exact Unlearning

Exact unlearning is implemented through retraining from scratch after removing the target subject from the training set. This model serves as the gold standard for subject-level unlearning, as it corresponds to the model that would have been obtained if the forgotten subject had never contributed training data. The exact retraining model uses the same classifier structure and hyperparameters as the baseline model, but is fitted only on the retain set. Using retraining without the forgotten data as the strongest reference point for evaluating approximate unlearning is consistent with prior unlearning literature [9, 11]. The exact retraining setting is important because it provides the strongest practical reference point for evaluating whether an approximate unlearning method behaves similarly to full removal.

3.8 Fuzzy Unlearning

Fuzzy unlearning is used as an approximate alternative to exact retraining. Instead of removing additional retained data beyond the forgotten subject, the method reduces the influence of retained training samples that are most similar to the forgotten subject in feature space. The goal is to weaken residual subject-specific structure while preserving as much retained-subject utility as possible. In the implementation, Subject 14 is selected as the target to forget. After feature extraction and splitting, the retained training set contains 1085 windows and the forgotten-subject training set contains 31 windows. The forgotten-subject test set contains 5 windows. These counts are important for interpreting the strength and stability of the reported forgetting results. Similarity is computed in the standardized feature space using a nearest-neighbor model fitted on the forgotten-subject training windows. For each retained training sample i , the k nearest forgotten-subject training samples are identified, where

$$k = \min(20, |\mathcal{D}_{\text{forget}}^{\text{train}}|).$$

In the present experiment, this gives $k = 20$ because the forgotten-subject training set contains 31 windows.

Let d_i denote the mean Euclidean distance from retained sample i to these k nearest forgotten-subject samples. A distance-based

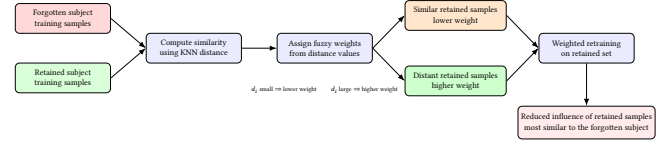


Figure 3: Workflow of the proposed fuzzy unlearning strategy. Similarity between retained and forgotten samples is computed in feature space and converted into fuzzy training weights. Retained samples that are more similar to the forgotten subject receive lower influence during weighted retraining.

fuzzy weight is then assigned as

$$w_i = 1 - \exp\left(-\frac{d_i^2}{2\sigma^2}\right),$$

where σ is set to the median of the distance values across the retained training set. The weights are clipped to the interval $[0, 1]$. Samples closer to the forgotten subject receive lower weights, while more distant samples retain greater influence during training.

The resulting model is still retrained from scratch on the retained set, but unlike exact retraining it uses similarity-based sample weights during fitting. In this sense, the method is best understood as a fuzzy-inspired, distance-based reweighting strategy rather than a full fuzzy inference system. It approximates forgetting by down-weighting retained samples that are closest to the forgotten subject instead of applying only hard exclusion.

3.9 Assessment Criteria

The experiments evaluate utility preservation, forgetting-related behavior, similarity between approximate and exact unlearning, and generalization. These criteria reflect the two central goals of unlearning: preserving performance on retained data while reducing information associated with the forgotten data.

3.9.1 Utility Preservation. For retained subjects, performance is measured using classification accuracy and macro-averaged F1 score on the retained-subject test set. Accuracy measures overall correctness, while macro F1 gives equal importance to each remaining subject class and is therefore more appropriate for multi-subject evaluation.

3.9.2 Forgetting-Related Measures. After unlearning, the forgotten subject is no longer a valid class for the exact and fuzzy models. Therefore, forgetting is evaluated through prediction behavior on forgotten-subject test windows rather than standard class accuracy alone. The following quantities are analyzed:

- mean maximum prediction confidence,
- predictive entropy,
- qualitative prediction distribution on forgotten-subject samples.

Lower confidence and higher entropy indicate that the model is less certain on forgotten-subject inputs and therefore less likely to retain strong subject-specific information.

3.9.3 Approximation to Exact Unlearning. Because fuzzy unlearning is intended to approximate exact retraining, the study also

measures agreement between the fuzzy and exact models. Two comparisons are used:

- prediction agreement, defined as the fraction of samples on which both models output the same class,
- mean absolute probability difference between the predictive distributions of the two models.

These comparisons are reported for both the retained-subject test set and the forgotten-subject test set. High agreement and small probability differences indicate that fuzzy unlearning behaves similarly to exact retraining.

3.9.4 Generalization Analysis. To assess overfitting, train and test performance are compared for all model variants. This is especially important in the present setting because EEG datasets are limited in size and sliding-window segmentation introduces dependence among nearby samples. As a result, very high training accuracy may overstate real generalization performance.

3.10 Experimental Workflow

Figure 4 summarizes the complete experimental workflow. First, the subject CSV files are loaded and assigned standard EEG channel names. Each recording is then split into training and test portions, segmented into sliding windows, and transformed into statistical and frequency feature vectors. Based on these features, retain-and-forget datasets are constructed for three model settings: a baseline classifier trained on all subjects, an exact unlearning model obtained by retraining without the forgotten subject, and a fuzzy unlearning model trained on the retain set with similarity-based sample weights. The final comparison is performed in terms of retained-subject utility, uncertainty on forgotten-subject samples, and generalization behavior.

This workflow allows the study of the privacy-utility trade-off of EEG subject unlearning in a controlled, feature-based setting. The emphasis on protecting privacy, retained utility, and computational effectiveness is also consistent with recent unlearning literature [9, 11].

4 Results

This section reports the empirical behavior of the three evaluated model settings: the baseline model trained on all subjects, the exact unlearning model retrained without the forgotten subject, and the fuzzy unlearning model obtained via similarity-based sample reweighting. The results are presented from two perspectives: retained-subject utility and behavior on forgotten-subject test samples.

4.1 Performance on Retained Subjects

Figure 5 compares classification performance on the retained-subject test set. Two metrics are reported: classification accuracy and macro-averaged F1 score. The baseline model obtains strong retained-subject performance, with an accuracy of 0.8971 and a macro F1 score of 0.8712. After removing the forgotten subject by exact retraining, the retained-subject performance remains high and even shows a slight improvement, reaching 0.9029 accuracy and 0.8998 macro F1. This indicates that removing the forgotten subject does not harm the classifier on the remaining classes and may even

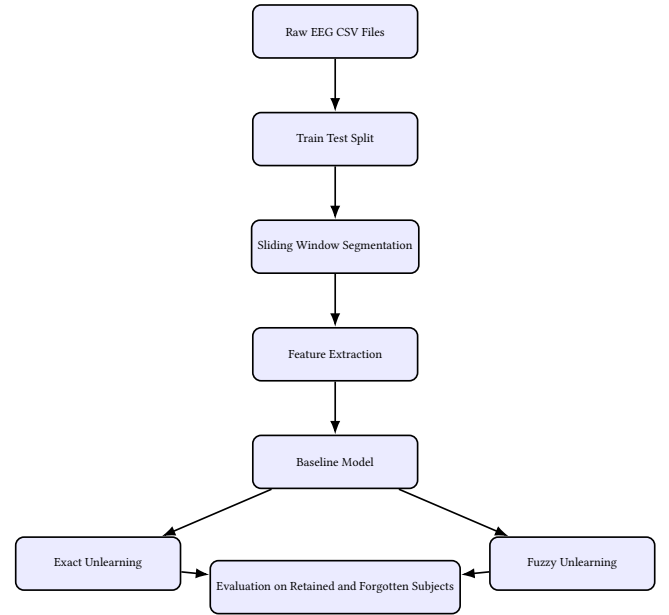


Figure 4: Simplified overview of the EEG unlearning pipeline.

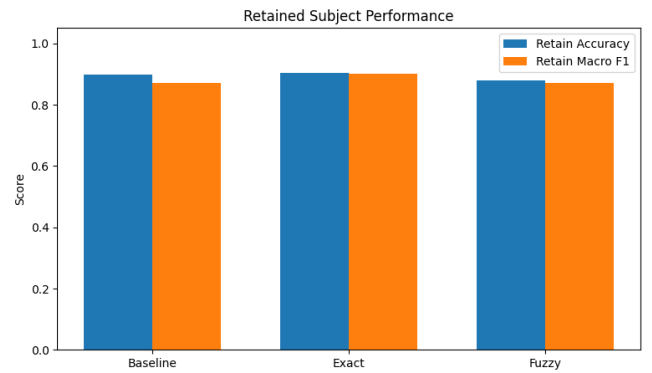


Figure 5: Retained-subject performance for the baseline, exact unlearning, and fuzzy unlearning models. Exact retraining achieves the best retained-set accuracy and macro F1, while fuzzy unlearning preserves most of the utility with a small decrease.

reduce some class interference. The fuzzy unlearning model also preserves a high level of retained-subject utility, with a smaller drop than exact retraining. Its retained-subject accuracy is 0.8800, and its macro F1 score is 0.8719. This result shows that the proposed fuzzy strategy maintains most of the predictive utility among the remaining subjects, although it does not match the retained-set performance of exact retraining. Overall, the retained-subject results suggest that both unlearning settings preserve useful discriminative structure for the remaining classes. Exact retraining gives the strongest retained-subject performance, while fuzzy unlearning secures competitive but slightly lower utility.

4.2 Behavior on Forgotten-Subject Test Samples

Figure 6 shows the behavior of the three models on test windows belonging to the forgotten subject. Since the forgotten subject is no longer a valid class after exact and fuzzy unlearning, forgetting is evaluated via predictive confidence and predictive entropy rather than standard class accuracy. The baseline model produces the highest confidence on forgotten-subject samples, with a mean maximum confidence of 0.3064 and an entropy of 2.9003. This indicates that the baseline model still retains subject-specific information and makes comparatively more certain predictions on samples from the subject that should later be forgotten. In contrast, the exact retraining model produces much lower confidence on forgotten-subject samples, with a mean maximum confidence of 0.1106, while the predictive entropy increases to 3.2650. This behavior is consistent with successful subject removal. The model becomes much less certain when presented with samples from the forgotten subject, suggesting that subject-specific information associated with that subject has been substantially reduced. The fuzzy unlearning model shows a very similar trend. Its mean maximum confidence on forgotten-subject samples is 0.1208, and its entropy is 3.2357. Compared with the baseline, this indicates a strong increase in uncertainty and a clear reduction in confidence in predictions about the forgotten subject. At the same time, the fuzzy model remains slightly more confident than the exact retraining model, suggesting that approximate unlearning weakens forgotten-subject information substantially but does not remove it as completely as exact retraining. Taken together, the results for the forgotten subject indicate that both exact and fuzzy unlearning move the model in the desired direction. The drop in prediction confidence and the increase in entropy indicate that the forgotten subject is no longer strongly represented in the classifier’s decision structure. Exact retraining remains the strongest reference for forgetting, while fuzzy unlearning approximates this behavior closely.

4.3 Comparison Between Exact and Fuzzy Unlearning

A direct comparison between the two unlearning settings reveals a clear trade-off. Exact retraining provides the strongest overall forgetting behavior and the best retained-subject performance in this experiment. Fuzzy unlearning, however, produces results that are close to exact retraining on the forgotten-subject evaluation while preserving most of the retained-subject utility. It is important to state the computational logic precisely. Fuzzy unlearning in the present implementation does not avoid retraining altogether. Instead, it retrains a new Random Forest model on the retained set while assigning lower sample weights to retained training windows that are most similar to the forgotten subject. The difference from exact retraining is therefore not the absence of retraining, but the replacement of hard similarity-independent treatment with similarity-based weighting. The empirical comparison supports this interpretation. On the retained-subject test set, the fuzzy and exact models reach a prediction agreement of 0.8914, with a mean absolute probability difference of 0.008498. On the forgotten-subject test set, prediction agreement reaches 1.0000, with a mean absolute probability difference of 0.010355. These values indicate that fuzzy unlearning reproduces the qualitative behavior of exact retraining

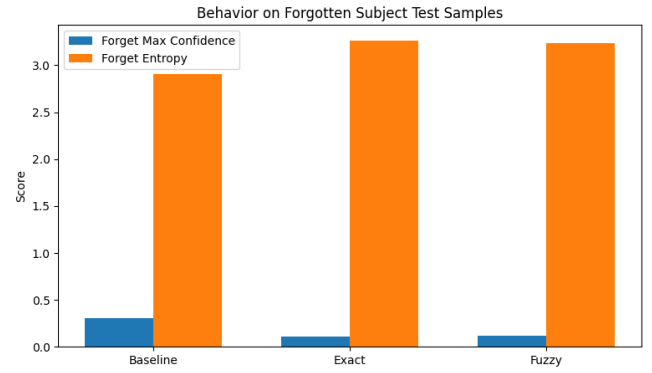


Figure 6: Behavior on forgotten-subject test samples. Compared with the baseline model, both exact retraining and fuzzy unlearning reduce the maximum prediction confidence and increase predictive entropy, indicating greater uncertainty about forgotten-subject inputs.

closely, especially on the forgotten-subject evaluation, while still showing a modest gap on retained-subject utility. In the present experiment, exact retraining achieves retained-subject accuracy and macro F1 of 0.9029 and 0.8998, while fuzzy unlearning reaches 0.8800 and 0.8719. On forgotten-subject test windows, exact retraining yields a mean maximum confidence of 0.1106 and entropy of 3.2650, whereas fuzzy unlearning yields 0.1208 and 3.2357. Thus, fuzzy unlearning moves strongly in the same direction as exact retraining, but does not fully match its retained-set performance or its forgetting strength.

4.4 Generalization and Overfitting Analysis

The train-versus-test comparison confirms that all three models operate in a setting with noticeable overfitting. The baseline model reaches a training accuracy of 0.9919 and training macro F1 of 0.9919, but its retained-subject test accuracy and macro F1 drop to 0.8971 and 0.8712. Exact retraining shows a similar pattern, with 0.9899 training accuracy and 0.9898 training macro F1, compared with 0.9029 and 0.8998 on the retained-subject test set. The fuzzy model reduces the training values slightly to 0.9751 accuracy and 0.9741 macro F1, but its retained-subject test performance remains lower at 0.8800 accuracy and 0.8719 macro F1. These results support the claim that limited dataset size and overlap between nearby windows make the task prone to optimistic training performance. The train-test gap should therefore be considered when interpreting both retained-subject utility and the apparent strength of subject identifiability in the baseline model.

4.5 Main Findings

The experimental results support three main observations. First, removing the forgotten subject does not reduce retained-subject performance, and exact retraining even yields the best results on the retained set. Second, both unlearning strategies significantly

increase predictive uncertainty on forgotten-subject samples compared with the baseline model. Third, fuzzy unlearning approximates the qualitative behavior of exact retraining while preserving most of the retained-subject utility.

These findings indicate that the proposed feature-based fuzzy unlearning pipeline can reduce the identifiability of removed subjects while preserving most of the model's utility among the retained subjects. At the same time, the results also confirm that exact retraining remains the strongest practical reference for complete subject removal.

4.6 Limitations

Several limitations should be considered when interpreting the results. First, all model variants show clear signs of overfitting. Training performance is extremely high, whereas test performance is consistently lower. This is partly expected because EEG data are high-dimensional, subject-dependent, and limited in size. In the present pipeline, the risk of overfitting is increased further by the use of handcrafted features and by repeated tuning of preprocessing and model settings during development. Second, the study relies on a single dataset source with 36 subjects recorded under one fixed experimental scenario. The findings should therefore be interpreted as evidence from a controlled proof-of-concept setting rather than as a conclusion about subject-level unlearning in EEG systems more generally. Third, the use of sliding-window segmentation introduces dependence among nearby samples. Even though the train and test segments are separated chronologically, neighboring windows from the same recording still share temporal structure. In the implementation, training windows are extracted with a step size of 1.5 seconds from 2-second windows, whereas test windows are non-overlapping. This design improves sample efficiency, but it also reduces the effective independence of the resulting dataset. Fourth, the forgotten-subject evaluation is based on a small number of forgotten test windows. In the current experiment, Subject 14 is the forgotten subject, with 31 forgotten training windows and only 5 forgotten test windows after segmentation. This is sufficient for a proof-of-concept comparison, but it limits the statistical strength of the conclusions about forgetting behavior. Fifth, the feature extraction stage itself is a limitation. The selected statistical and spectral band-power features provide an interpretable and computationally practical representation, but they cannot capture all subject-specific EEG patterns relevant to forgetting. At the same time, manually designed features may preserve redundant or correlated information, which can contribute to both overfitting and incomplete removal of forgotten-subject characteristics. Overall, the project should be understood as an exploratory feature-based study of EEG subject unlearning under practical data constraints. Future work should evaluate the approach on larger and more diverse datasets, test subject-independent or cross-session settings, and investigate stronger regularization or alternative representations to reduce overfitting more systematically.

5 Conclusion

This work investigated subject-level unlearning in an EEG classification setting from a privacy and security perspective. The objective was to examine whether subject-related information associated

with one selected user could be reduced without causing a major loss in classification performance for the remaining subjects. To study this question, three settings were compared: a baseline model trained on all subjects, an exact unlearning model obtained by retraining without the forgotten subject, and a fuzzy-inspired unlearning model obtained by distance-based reweighting of retained training samples. The results show a consistent pattern. The baseline model retains strong information about the forgotten subject, achieving a forgotten-subject prediction rate of 1.0000 for Subject 14 and a mean maximum confidence of 0.3064 on forgotten-subject test windows. Exact retraining produces the strongest forgetting behavior, reducing mean maximum confidence to 0.1106 and increasing predictive entropy to 3.2650, while also achieving the best retained-subject performance with 0.9029 accuracy and 0.8998 macro F1. Fuzzy unlearning follows the same qualitative trend. It reduces mean maximum confidence on forgotten-subject test windows to 0.1208 and increases entropy to 3.2357, while preserving most retained-subject utility with 0.8800 accuracy and 0.8719 macro F1. A direct comparison with exact retraining shows that fuzzy unlearning approximates the exact model closely. Prediction agreement between the fuzzy and exact models reaches 0.8914 on the retained-subject test set and 1.0000 on the forgotten-subject test set, with small mean absolute probability differences in both cases. At the same time, the train-test analysis confirms that all variants suffer from overfitting, reflecting the limited dataset size and the dependence introduced by sliding-window segmentation. Taken together, these results support the view that distance-based fuzzy reweighting is a viable approximate unlearning strategy in this feature-based EEG setting. It does not replace exact retraining as the strongest reference for subject removal. Still, it moves the model in the desired privacy-preserving direction while preserving most of the retained-subject utility.

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